PROBLEM 18

Heavy Particle on a Cone

Original statement · Problems.pdf, p. 10

Conical Surface

A heavy particle of mass m is moving without friction on a cone which is described as:

$$z^2 = x^2 + y^2$$

The frame of reference $Oxyz$ is inertial and Oz is the ascending vertical. The initial coordinates of the particle are $(a, 0, a)$ and its velocity is $\sqrt{ag}\,\mathbf{j}$. Reduce the problem to integral equations and determine the maximum and minumum [sic] value of z along the motion.

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** a heavy particle of mass m moves without friction on the interior of the cone $z^2 = x^2 + y^2$, $z > 0$. The frame $Oxyz$ is inertial and Oz is the ascending vertical.
- **Constraint:** the particle stays on the surface, $\Phi(x, y, z) = x^2 + y^2 - z^2 = 0$ (holonomic, scleronomic). It is unilateral in principle (the particle can leave the cone if the reaction would have to pull it), so we verify $N \geq 0$ at the end. The surface is smooth, so the reaction is normal to it.
- **Degrees of freedom:** $\mathrm{CDOF} = 3 - 1 = 2$. We choose the cylindrical coordinates (r, θ) and, on the cone, $z = r$.
- **Initial state:** position $(a, 0, a)$, i.e. $r(0) = a, \theta(0) = 0, z(0) = a$; velocity $\sqrt{ag}\,\mathbf{j}$, i.e. $\dot{r}(0) = 0, r(0)\dot{\theta}(0) = \sqrt{ag}, \dot{z}(0) = 0$.

- **Parameters:** m [kg], g [m/s^2], a [m].

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ with $\mathbf{k}$ vertical upward.
- **Cylindrical basis:** $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$, the inertial basis rotated by $\theta(t)$ about $\mathbf{k}$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\theta + \sin^2\theta = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

Multiplying out, RR^T has diagonal entries $\cos^2 + \sin^2 = 1$ of the rotation angle (and 1) and off-diagonal entries $\pm(\cos\sin - \sin\cos) = 0$, so it is the identity.

- The angular velocity of $\mathcal{B}_1$ is $\boldsymbol{\omega}_{10} = \dot\theta\,\mathbf{k}$, so Poisson's formula gives $\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_r$.
- **Surface-adapted basis.** On the cone $z = r$ (slope 45°) the unit generator is $\mathbf{e}_g = (\mathbf{e}_r + \mathbf{k})/\sqrt{2}$ and the unit normal pointing into the cone, where the particle lies, is $\mathbf{n}_{in} = (\mathbf{k} - \mathbf{e}_r)/\sqrt{2}$. The triple $\{\mathbf{e}_g, \mathbf{e}_\theta, \mathbf{n}_{in}\}$ is right-handed ($\mathbf{e}_g \times \mathbf{e}_\theta = \mathbf{n}_{in}$) and its components in $\mathcal{B}_1$ are the columns of

$$[{}_1R_2] = \begin{pmatrix} 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \end{pmatrix}, \qquad \det[{}_1R_2] = \tfrac{1}{2} + \tfrac{1}{2} = 1, \qquad [{}_1R_2][{}_1R_2]^T = I$$

PHASE 3

Derivation

3.1 Kinematics

Pedagogical Justification: the position is written with the cylindrical coordinates and the constraint $z = r$ is applied before differentiating, so that only the free coordinates (r, θ) remain. The derivatives of $\mathbf{e}_r$ use Poisson's formula.

$$\mathbf{r} = r\,\mathbf{e}_r + r\,\mathbf{k}, \qquad \mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\theta\,\mathbf{e}_\theta + \dot r\,\mathbf{k}, \qquad v^2 = 2\dot r^2 + r^2\dot\theta^2$$

$$\mathbf{a} = \left(\ddot r - r\dot\theta^2\right)\mathbf{e}_r + \left(r\ddot\theta + 2\dot r\dot\theta\right)\mathbf{e}_\theta + \ddot r\,\mathbf{k}$$

3.2 Forces

- **Applied force:** the weight $-mg\,\mathbf{k}$ (conservative, $V = mgz = mgr$).
- **Constraint force:** the reaction of the smooth cone, normal to the surface, i.e. along $\nabla\Phi \propto \mathbf{e}_r - \mathbf{k}$. With an unknown magnitude ν (positive when pressing the particle against the surface from inside), $\mathbf{N} = \nu\,\mathbf{n}_{in} = \dfrac{\nu}{\sqrt{2}}\,(\mathbf{k} - \mathbf{e}_r)$. It has no $\mathbf{e}_\theta$ component.
- **Inertia forces:** none (S_0 is inertial).

3.3 Equations of Motion

> **Pedagogical Justification:** Newton's second law in S_0 projected on the cylindrical basis gives three scalar equations for the three unknowns r, θ and ν (the constraint $z = r$ is already used):

$$\mathbf{e}_r:\ m\left(\ddot r - r\dot\theta^2\right) = -\frac{\nu}{\sqrt{2}}, \qquad \mathbf{e}_\theta:\ m\left(r\ddot\theta + 2\dot r\dot\theta\right) = 0, \qquad \mathbf{k}:\ m\ddot r = \frac{\nu}{\sqrt{2}} - mg$$

- **Dynamical decoupling.** Adding the $\mathbf{e}_r$ and $\mathbf{k}$ equations eliminates the reaction, which is then obtained from the last equation once $r(t)$ is known: $2\ddot r - r\dot\theta^2 = -g$.
- **First integral from $\mathbf{e}_\theta$.** The azimuthal acceleration is $r\ddot\theta + 2\dot r\dot\theta = \frac{1}{r}\frac{d}{dt}\left(r^2\dot\theta\right)$ (product rule), so $r^2\dot\theta$ is constant:

$$\frac{d}{dt}\left(r^2\dot\theta\right) = 0 \implies r^2\dot\theta = r(0)^2\dot\theta(0) = a\sqrt{ag} \implies \dot\theta = \frac{a\sqrt{ga}}{r^2}$$

- **Energy.** The reaction does no work ($\mathbf{N}\cdot\mathbf{v} = 0$ because $\mathbf{N}$ is normal to the surface and $\mathbf{v}$ is tangent to it) and gravity is conservative, so $E = \frac{1}{2}mv^2 + mgz$ is constant. This is a first integral of the equations above and replaces the radial equation:

$$E = \tfrac{1}{2}m\left(2\dot{r}^2 + r^2\dot{\theta}^2\right) + mgr = \tfrac{1}{2}m(ag) + mga = \tfrac{3}{2}mga$$

3.4 Integration and Answers

Reduction to quadratures

> **Pedagogical Justification:** substitute $\dot{\theta} = a\sqrt{ga}/r^2$ in the energy, so that only r and $\dot{r}$ remain, and isolate $\dot{r}^2$:

$$m\dot{r}^2 + \frac{mga^3}{2r^2} + mgr = \frac{3}{2}mga \implies \dot{r}^2 = \frac{3}{2}ga - gr - \frac{ga^3}{2r^2} = \frac{g}{2r^2}\left(3ar^2 - 2r^3 - a^3\right)$$

- **General quadrature.** For arbitrary initial data, with energy E and $h = r^2\dot{\theta}$, the same steps give $m\dot{r}^2 + \dfrac{mh^2}{2r^2} + mgr = E$, and separating variables with Barrow's rule from $r(0) = r_0$:

$$\int_{r_0}^{r} \frac{dr'}{\sqrt{\dfrac{E - mgr'}{m} - \dfrac{h^2}{2r'^2}}} = \pm\int_0^t dt' = \pm t, \qquad \theta(t) = \theta_0 + \int_0^t \frac{h}{r(t')^2}\,dt'$$

- **These initial data** ($E = \frac{3}{2}mga$, $h = a\sqrt{ga}$, $r_0 = a$) give the radicand $\dfrac{g}{2r^2}\left(-2r^3 + 3ar^2 - a^3\right)$, which is shown below to be ≤ 0 and to vanish only at $r = a$; the separation of variables is then not possible ($\dot{r} \equiv 0$) and the motion is found directly:

Maximum and minimum of $z = r$

> **Pedagogical Justification:** the turning points of r are the roots of $\dot{r}^2 = 0$, i.e. of $P(r) = -2r^3 + 3ar^2 - a^3$. Since the particle starts at $r = a$ with $\dot{r}(0) = 0$, $r = a$ is a root: $2a^3 - 3a^3 + a^3 = 0$. Dividing $2r^3 - 3ar^2 + a^3$ by $(r - a)$ (check: $(r-a)(2r^2 - ar - a^2) = 2r^3 - ar^2 - a^2r - 2ar^2 + a^2r + a^3 = 2r^3 - 3ar^2 + a^3$):

$$2r^3 - 3ar^2 + a^3 = (r - a)\left(2r^2 - ar - a^2\right) = (r-a)(r-a)(2r + a) = (r-a)^2(2r + a)$$

- Therefore

$$\dot{r}^2 = -\frac{g\,(2r+a)}{2r^2}\,(r-a)^2$$

For $r > 0$ the factor $(2r + a)$ is positive, so the right-hand side is ≤ 0, and it vanishes only at $r = a$. Since $\dot{r}^2 \geq 0$, the only possible motion is

$$\boxed{r(t) \equiv a, \qquad z_{\max} = z_{\min} = a}$$

The data of the problem are exactly those of a circular orbit at constant height.

PHASE 4

Interpretation & Checks

- **Steady circular orbit:** the initial speed $\sqrt{ag}$ is the circular-orbit speed on the cone, $v^2 = gr\tan\gamma$ with $\tan\gamma = dz/dr = 1$. The particle sits at the minimum of the effective potential $V_{\text{eff}}(r) = mgr + \dfrac{ma^3 g}{2r^2}$ and describes a horizontal circle with $\dot{\theta} = \sqrt{g/a}$.
- **Reaction on the orbit.** With $r = a$ and $\dot{\theta}^2 = g/a$ the acceleration is $\mathbf{a} = -g\,\mathbf{e}_r$. From $\mathbf{N} - mg\,\mathbf{k} = m\mathbf{a}$: $\mathbf{N} = mg\,(\mathbf{k} - \mathbf{e}_r) = \sqrt{2}\,mg\,\mathbf{n}_{in}$, so $\nu = \sqrt{2}\,mg > 0$: the cone pushes the particle, and the unilateral constraint stays active.
- **Stability.** The effective inertia of the r-motion is $2m$ (the kinetic energy has $2\dot{r}^2$) and $V''_{\text{eff}}(a) = 3mg/a$, hence $\omega_r^2 = 3g/(2a)$: small radial oscillations have period $2\pi/\sqrt{3g/(2a)}$.
- **Dimensional check:** $[r^2\dot{\theta}] = \mathrm{m^2 s^{-1}}$ and $[ga^3/r^2] = \mathrm{m^2 s^{-2}}$, like $[\dot{r}^2]$, so every term of the energy equation has the same units.

- **Check against the official key:** `Problems.pdf` has no entry for Problem 18 in its solution keys, so the result is verified by independent derivation. Sympy confirms $\dot{r}^2 = -g(a-r)^2(a+2r)/(2r^2)$ and $\mathbf{N} = mg(\mathbf{k} - \mathbf{e}_r)$ with $\|\mathbf{N}\| = \sqrt{2}\,mg$. Numerical integration ($g = a = 1$) keeps $r = 1$ to 10^{-12}, and for a perturbed start $r(0) = 1.001$ the radial period is 5.13020, against $2\pi/\sqrt{3/2} = 5.13020$.

Source: Problems.pdf